

Rufidatul Radium

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Full-stack Software Engineer with experience in backend systems, AI integrations, and scalable web applications. Skilled in FastAPI, Next.js, and distributed systems, with hands-on experience in LLM workflows, automation pipelines, and real-time applications. Strong background in machine learning, computer vision, and API integrations.

Experience

Nymph Solutions Ltd. (Remote)

May 2026 – Present

Full Stack Engineer

- Built a **social and messaging channel integration from scratch**, implementing OAuth2 authentication, real-time webhooks, direct messaging, media handling, rate-limited polling, and reliable event synchronization for a customer engagement platform
- Developed an **AI auto-reply pipeline** with duplicate-message detection, conversation-context grounding, and safeguards against irrelevant or repeated responses
- Diagnosed and resolved a **production-wide real-time messaging outage** caused by a network-level pub/sub failure, and fixed mobile reconnect issues affecting message delivery
- Optimized a high-traffic comment feed from approximately **1,300 database queries to 23 queries per load**, significantly reducing database overhead and improving response performance
- Delivered full-stack features using **FastAPI and React** for a fleet and rental management SaaS, including revenue ledgers with PDF export, booking lifecycle management, and availability scheduling
- Built a companion **CMS plugin** to synchronize fleet data with a public-facing storefront, enabling consistent inventory and booking information across systems
- Designed and deployed an **image-similarity search service** using embedding models and vector search, reducing catalogue re-indexing time from approximately **31 minutes to 5 seconds**
- Extended a **multi-provider AI engine** with LLM provider integrations, visual search, duplicate-reply detection, and automated follow-up messaging for an AI-powered e-commerce chat application
- Led a major **upstream merge and codebase reconciliation effort**, resolving divergence between a long-lived product fork and its open-source base while preserving product-specific functionality

Singularity Limited (Remote)

Jun 2025 – May 2026

Backend Engineer

- Architected and extended a customized **Chatwoot-based customer engagement platform**, enabling unified communication across Facebook, Instagram, WhatsApp, Telegram, Tiktok and Shopify
- Designed and implemented **event-driven automation systems** for posts, comments, and messages, introducing granular event types (post/comment/message lifecycle) to power workflow automation
- Built **AI-powered response pipelines** using LLM integrations (OpenAI/OpenRouter), including prompt orchestration, media-aware context injection, and fallback handling for non-user (activity/system) events
- Developed and maintained **webhook processing systems** for Meta (Facebook/Instagram) and WhatsApp APIs, handling high-volume real-time events with Sidekiq background jobs and ActionCable broadcasting
- Engineered **message processing logic** to accurately distinguish between user-generated, system-generated, and activity events, improving AI response accuracy and preventing incorrect triggers
- Implemented **contact intelligence pipelines** to extract emails and multi-country phone numbers from conversations, normalize to E.164 format, validate inputs, and persist structured metadata with source attribution
- Built **resilient media handling pipelines**, including image ingestion, storage via ActiveStorage (S3-compatible), and AI-based image understanding with optimized payload handling
- Optimized system performance using **Sidekiq queues, Redis tuning, and background job orchestration**, ensuring reliability under high message throughput
- Debugged and resolved complex production issues involving **webhooks, API failures, queue backlogs, and real-time messaging inconsistencies**
- Integrated external platforms including **Shopify and WooCommerce**, enabling product/order data access for AI-driven workflows and automation tools
- Containerized services using **Docker and Docker Compose** for consistent development and deployment environments
- Contributed to **scalable backend architecture** with PostgreSQL, including vector database usage for semantic search and AI context retrieval

- Contributed to **Banglar Math**, a distributed educational platform for Bangladeshi students
- Built **admin dashboard** using Next.js by translating Figma designs into responsive components
- Integrated **REST APIs** using Swagger and implemented pagination, improving content management workflows
- Implemented **form validation** using React Hook Form and Zod for robust input handling

- Built an **OCR-based NID data extraction pipeline** supporting Bangla and English using OpenCV, Tesseract, and EasyOCR; implemented preprocessing, card-side detection, and face validation with Dlib
- Automated **KYC and underwriting workflows** using UiPath and Apache NiFi, reducing manual data entry and improving processing efficiency
- Developed **predictive models** (Gradient Boosting, Decision Trees) for lien underwriting, achieving 82% accuracy in decision automation
- Designed and implemented a **secure licensing module** for Insursoft using Spring Boot, Angular, and Oracle DB with SHA-based validation and controlled database access

Projects

Kitchen Buddy (LLM + RAG System)

- Built a mobile-first recipe recommendation system using **React Native** and **FastAPI**
- Implemented **Retrieval-Augmented Generation (RAG)** pipeline to generate context-aware recipe suggestions based on available ingredients
- Designed backend APIs for ingredient management, recipe retrieval, and LLM interaction
- Integrated structured data storage with SQLite and optimized prompt workflows for relevant responses

Ekunjo – Urban Farming Platform

- Developed a full-stack platform using **Node.js, React, MongoDB, and FastAPI** to support urban farming
- Built features including **plant guides, fertilizer recommendations, and nursery location mapping** using Leaflet
- Integrated **AI-based plant disease detection** via image upload and prediction API
- Implemented **real-time chat system** using Firebase Firestore and live listeners

WebInsight Chrome Extension

- Built a Chrome extension to **analyze active browser tabs** and generate summaries and answers
- Implemented dynamic **tab content extraction** and processing pipeline
- Integrated backend LLM APIs for contextual Q&A and multi-tab summarization

Travel Tales

- Developed a travel planning platform using **Next.js, NestJS, and FastAPI**
- Integrated **Map and Weather APIs** for location-based trip planning
- Implemented LLM-powered features for generating travel suggestions and itineraries

Skills

- **Languages:** JavaScript, Python, Java, C
- **Frameworks:** Next.js, React, Node.js, FastAPI, Spring Boot
- **AI/ML:** NLP, Computer Vision, Transformers, RAG, Prompt Engineering
- **Database:** MongoDB, MySQL, PostgreSQL
- **Tools:** Docker, Git, Postman, UiPath, Apache NiFi

Education

Achievements

- Hackathon Finalist (OpenAPI), ITVerse 2023
- TechTales Finalist, ITVerse 2023
- Competitive Programming Finalist, NGPC 2021